

First Order Differential Equations (Chapter 2)

Separable Equation: A first-order differential equation of the form

$$\frac{dy}{dx} = g(x)h(y)$$

is said to be **separable** or to have **separable variables**. For example, $\frac{dy}{dx} = xy^2e^{3x+4y}$ is a **separable DE** and $\frac{dy}{dx} = y + \sin x$ is a **non-separable DE**.

Chapter 2.2: Example(1-4)

Exercise 2.2: 6, 7, 11-14, 24, 25

Linear Equation: A first-order differential equation of the form

$$a_1(x)\frac{dy}{dx} + a_0(x)y = g(x)$$

is said to be a linear equation in the variable y .

Standard Form: By dividing both sides of the above equation by the lead coefficient we obtain a more useful form, the **standard form**, of a linear equation:

$$\frac{dy}{dx} + P(x)y = f(x)$$

For example, $\frac{dy}{dx} + 2xy = 0$, $\frac{dy}{dx} = y + 5$ are both linear and separable, but that the linear equation $\frac{dy}{dx} + y = x$ is not separable

SOLVING A LINEAR FIRST-ORDER EQUATION

- (i) Remember to put a linear equation into the standard form (2).
- (ii) From the standard form of the equation identify $P(x)$ and then find the integrating factor $e^{\int P(x)dx}$. No constant need be used in evaluating the indefinite integral $\int P(x)dx$.
- (iii) Multiply the both sides of the standard form equation by the integrating factor. The left-hand side of the resulting equation is automatically the derivative of the product of the integrating factor $e^{\int P(x)dx}$ and y :

$$\frac{d}{dx} [e^{\int P(x)dx} y] = e^{\int P(x)dx} f(x).$$

- (iv) Integrate both sides of the last equation and solve for y .

Chapter 2.3: Example(1-4, 6)

Exercise 2.3: 9, 12, 19, 24, 39, 40

Differential of a Function of Two Variables: If $z = f(x, y)$ is a function of two variables with continuous first partial derivatives in a region R of the xy -plane, then its differential is

$$dz = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$$

In the special case when $f(x, y) = c$, where c is a constant, then above equation becomes,

$$\frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = 0$$

In other words, given a one-parameter family of functions $f(x, y) = c$, we can generate a first-order differential equation by computing the differential of both sides of the equality.

Of course, not every first-order DE written in differential form

$M(x, y)dx + N(x, y)dy = 0$ corresponds to a differential of $f(x, y) = c$.

Exact Equation: A differential expression $M(x, y)dx + N(x, y)dy$ is an **exact differential** in a region R of the xy -plane if it corresponds to the differential of some function $f(x, y)$ defined in R . A first-order differential equation of the form $M(x, y)dx + N(x, y)dy = 0$ is said to be an **exact equation** if the expression on the left-hand side is an exact differential.

Theorem: Criterion for an Exact Differential: Let $M(x, y)$ and $N(x, y)$ be continuous and have continuous first partial derivatives in a rectangular region R defined by $a < x < b, c < y < d$. Then a necessary and sufficient condition that $M(x, y)dx + N(x, y) dy$ be an exact differential is

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Method of Solution Given an equation in the differential form $M(x, y) dx + N(x, y) dy = 0$, determine whether the equation is exact or not. If it is exact DE then there exists a function f for which

$$\frac{\partial f}{\partial x} = M(x, y).$$

We can find f by integrating $M(x, y)$ with respect to x while holding y constant:

$$f(x, y) = \int M(x, y) dx + g(y), \quad (1)$$

where the arbitrary function $g(y)$ is the “constant” of integration. Now differentiate (1) with respect to y and assume that $\partial f / \partial y = N(x, y)$:

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y} \int M(x, y) dx + g'(y) = N(x, y).$$

This gives
$$g'(y) = N(x, y) - \frac{\partial}{\partial y} \int M(x, y) dx. \quad (2)$$

Finally, integrate (2) with respect to y and substitute the result in (1). The implicit solution of the equation is $f(x, y) = c$.

Integrating Factors: We already learn from the previous chapter that the left-hand side of the linear equation $y' + P(x)y = f(x)$ can be transformed into a derivative when we multiply the equation by an integrating factor. The same basic idea sometimes works for a nonexact differential equation $M(x, y)dx + N(x, y)dy = 0$. That is, it is sometimes possible to find an **integrating factor** $\mu(x, y)$ so that after multiplying, the left-hand side of $\mu(x, y)M(x, y)dx + \mu(x, y)N(x, y)dy = 0$ is an exact differential.

Criterion for converting a **nonexact DE** to an **exact DE**:

We summarize the results for the differential equation

$$M(x, y) dx + N(x, y) dy = 0. \quad (1)$$

- If $(M_y - N_x)/N$ is a function of x alone, then an integrating factor for (1) is

$$\mu(x) = e^{\int \frac{M_y - N_x}{N} dx}.$$

- If $(N_x - M_y)/M$ is a function of y alone, then an integrating factor for (1) is

$$\mu(y) = e^{\int \frac{N_x - M_y}{M} dy}.$$

Chapter 2.4: Example(1-4) Exercise 2.4: 4, 5, 7, 9, 12, 13, 21, 22, 27-30, 32, 35-38

 **Substitutions** Often the first step in solving a differential equation consists of transforming it into another differential equation by means of a **substitution**. For example, suppose we wish to transform the first-order differential equation $dy/dx = f(x, y)$ by the substitution $y = g(x, u)$, where u is regarded as a function of the variable x . If g possesses first-partial derivatives, then the Chain Rule

$$\frac{dy}{dx} = \frac{\partial g}{\partial x} \frac{dx}{dx} + \frac{\partial g}{\partial u} \frac{du}{dx} \quad \text{gives} \quad \frac{dy}{dx} = g_x(x, u) + g_u(x, u) \frac{du}{dx}.$$

If we replace dy/dx by the foregoing derivative and replace y in $f(x, y)$ by $g(x, u)$, then the DE $dy/dx = f(x, y)$ becomes $g_x(x, u) + g_u(x, u) \frac{du}{dx} = f(x, g(x, u))$, which, solved for du/dx , has the form $\frac{du}{dx} = F(x, u)$. If we can determine a solution $u = \phi(x)$ of this last equation, then a solution of the original differential equation is $y = g(x, \phi(x))$.

Homogeneous Equations If a function f possesses the property $f(tx, ty) = t^\alpha f(x, y)$ for some real number α , then f is said to be a **homogeneous function** of degree α . For example, $f(x, y) = x^3 + y^3$ is a homogeneous function of degree 3, since

$$f(tx, ty) = (tx)^3 + (ty)^3 = t^3(x^3 + y^3) = t^3 f(x, y),$$

whereas $f(x, y) = x^3 + y^3 + 1$ is not homogeneous. A first-order DE in differential form

$$M(x, y) dx + N(x, y) dy = 0$$

is said to be **homogeneous*** if both coefficient functions M and N are homogeneous functions of the *same* degree.

- ❖ The substitutions that can be used to solve a homogeneous differential equation. Specifically, either of the substitutions $y = ux$ or $x = vy$, where u and v are new dependent variables, will reduce a **homogeneous** equation to a **separable** first-order differential equation.

Bernoulli's Equation The differential equation

$$\frac{dy}{dx} + P(x)y = f(x)y^n, \quad (1)$$

where n is any real number, is called **Bernoulli's equation**. Note that for $n = 0$ and $n = 1$, equation (1) is linear. For $n \neq 0$ and $n \neq 1$ the substitution $u = y^{1-n}$ reduces any equation of form (1) to a linear equation.

Reduction to Separation of Variables A differential equation of the form

$$\frac{dy}{dx} = f(Ax + By + C)$$

can always be reduced to an equation with separable variables by means of the substitution $u = Ax + By + C, B \neq 0$.